

ON THE SIGN OF INTER-CLASS INTERFERENCE IN DIRICHLET PARTIAL SUMS: A PEAK–TROUGH RECONCILIATION

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ABSTRACT. The ω -class decomposition of a Dirichlet partial sum $D_F(t; N) = \sum_{n \leq N} a_n n^{-1/2-it}$ partitions it into sub-sums S_k indexed by the number of distinct prime factors $\omega(n) = k$, and the *inter-class energy ratio* $r(t; N) = (|D_F|^2 - \sum_k |S_k|^2) / \sum_k |S_k|^2$ measures whether the classes interfere constructively ($r > 0$) or destructively ($r < 0$). Two lines of computational investigation within a single research program have reported *opposite* signs for this quantity: an early phase reported a representative *destructive* value $r \approx -0.82$ (interpreted as characteristic of the Riemann zeta function), while a later, canonical pipeline found r to be strongly *constructive* at the largest values of $|D_F|$. We resolve the apparent contradiction. The resolution is structural rather than numerical: by the mean value theorem for Dirichlet polynomials the *unconditional* average of r vanishes, so r carries information only *conditionally*, and its sign is governed entirely by the event on which one conditions. We show that r is a monotone increasing function of the conditioning quantile of $|D_F|$: it is negative at *troughs* (local minima of $|D_F|$) and positive at *peaks* (local maxima), crossing zero in the bulk. The frequently cited value $r \approx -0.82$ was measured at $t = 150000$, which we identify as a trough of $|D_\zeta(t; N)|$, not a peak. Both prior reports are therefore correct about their respective conditioning sets; we recommend that all inter-class interference statements specify the conditioning explicitly. A fully reproducible Google Colab notebook accompanies the note.

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1. INTRODUCTION

Decomposing a Dirichlet partial sum by the number of distinct prime factors of its index—the ω -class decomposition—has proved a useful lens on how large values of L -functions are assembled from their arithmetic building blocks. Writing

$$(1) \quad D_F(t; N) = \sum_{n \leq N} \frac{a_n}{n^{1/2+it}} = \sum_{k \geq 0} S_k(t; N), \quad S_k(t; N) = \sum_{\substack{n \leq N \\ \omega(n)=k}} \frac{a_n}{n^{1/2+it}},$$

one asks how the class-sums S_k combine: do their phases reinforce (constructive interference) or cancel (destructive interference)? The natural scalar summarizing this is the inter-class energy ratio $r(t; N)$ of Definition 2.2 below.

A tension has appeared in the computational record of this program. An early phase, calibrating against the Davenport–Heilbronn function and sampling D_ζ near $t = 1.5 \times 10^5$, reported a representative *negative* value $r \approx -0.82$ and described the inter-class interference of ζ as predominantly *destructive*—a “cancellation architecture” among composite classes. A later phase, using a canonical Parseval-validated pipeline and conditioning on the largest values of $|D_\zeta|$, found the opposite: at the highest peaks r is strongly *positive* (constructive), with mean $r \approx +1.5$ and more than 90% of peaks satisfying $r > 0$. Taken at face value the two reports contradict one another.

The purpose of this note is to show that they do not. The contradiction dissolves once one recognizes that r is an intrinsically *conditional* quantity (Section 3): its unconditional mean is zero, so any nonzero value is an artifact of the conditioning event, and *the sign of r is a function of which event one selects*. We establish (Section 4) that r increases monotonically with the conditioning quantile of $|D_F|$, passing from negative at troughs to positive at peaks. In Section 5 we show that the often-cited point $t = 150000$ is a trough, which is exactly why it returns a negative r , and we reconcile the two reports. Section 6 states the recommended convention for the corpus. Appendix A provides a self-contained, reproducible computation.

2. SETUP

Definition 2.1 (ω -class sum). For $F(s) = \sum a_n n^{-s}$ and $k \geq 0$, the ω -class partial sum is $S_k(t; N)$ as in (1), where $\omega(n)$ is the number of distinct prime factors of n (with $\omega(1) = 0$). Thus $D_F = \sum_k S_k$.

Definition 2.2 (Inter-class energy ratio). Define the *class energy* $E(t) = \sum_k |S_k(t)|^2$, the *cross-term* $X(t) = |D_F(t)|^2 - E(t) = \sum_{j \neq k} \operatorname{Re}[S_j \overline{S_k}]$, and the *inter-class energy ratio*

$$(2) \quad r(t; N) = \frac{X(t)}{E(t)}.$$

From the identity $|D_F|^2 = E(1 + r)$: $r > 0$ is constructive interference between classes, $r < 0$ destructive, $r = 0$ independent.

We take $F = \zeta$ ($a_n = 1$) throughout, with a random multiplicative control (coefficients $a_p = e^{i\theta_p}$, θ_p i.i.d. uniform, extended multiplicatively) for comparison. All sums use Kahan compensated summation; the t -grid uses spacing $\Delta t = 2\pi/\log N$; $D_\zeta(t; 10^4)$ agrees with a 50-digit `mpmath` reference to ≥ 10 digits.

3. r IS A CONDITIONAL QUANTITY

The single fact that organizes everything is that r has no unconditional content.

Proposition 3.1 (Unconditional vanishing). *By the Montgomery–Vaughan mean value theorem for Dirichlet polynomials, for $j \neq k$,*

$$(3) \quad \frac{1}{T} \int_0^T \operatorname{Re}[S_j(t) \overline{S_k(t)}] dt = O\left(\frac{N}{T}\right) \quad (T \rightarrow \infty),$$

and hence $\frac{1}{T} \int_0^T X(t) dt = O(N/T)$. Consequently the unconditional average of the cross-term vanishes, and any persistent nonzero value of r is a property of a conditional ensemble, not of D_F as a whole.

The proposition has an immediate and decisive corollary for the present question.

Corollary 3.2. *The sign of r is not an intrinsic attribute of F . It is defined only relative to a conditioning event $\mathcal{E} \subseteq [T, 2T]$, as $\langle r \rangle_{\mathcal{E}} = \mathbb{E}[r(t) \mid t \in \mathcal{E}]$. Different $\mathcal{E}$ may—and, as we show, do—yield opposite signs.*

Because the bulk average is zero while individual values of r are $O(1)$, the distribution of r over t is two-sided with mean near zero. Selecting the high- $|D_F|$ tail samples one side of this distribution; selecting the low- $|D_F|$ tail samples the other. The remainder of the note makes this quantitative.

4. MAIN RESULT: r INCREASES WITH THE CONDITIONING QUANTILE

Fix N and a window $[T, 2T]$. For $q \in [0, 1]$ let Q_q denote the event “ $|D_F(t; N)|$ lies in the q -th quantile band of its distribution over the window.” Write $\bar{r}(q) = \langle r \rangle_{Q_q}$ for the conditional mean of r on that band.

Theorem 4.1 (Peak–trough sign reversal; empirical). *For $F = \zeta$ at $N = 10^5$, sampled over 5498 abscissae with spacing $\Delta t = 2\pi/\log N$, the decile-conditional mean $\bar{r}(q)$ (Table 1, Figure 1) is a strictly monotone increasing function of q :*

- (i) **Troughs** (lowest decile of $|D_{\zeta}|$): every sampled point has $r < 0$ (fraction 1.00), with mean $\bar{r} = -0.998$.
- (ii) **Bulk**: $\bar{r}(q)$ crosses zero near $q \approx 0.69$.
- (iii) **Peaks** (top decile): $\bar{r} = +1.544$, with 92% of points satisfying $r > 0$; both the fraction and the magnitude rise further for the most extreme peaks.

The random multiplicative control (mean of five seeds $\{42, 137, 271, 314, 577\}$) exhibits the same profile, with peak mean $\bar{r} = +1.41 \pm 0.04$ and the same zero-crossing near $q \approx 0.65$. The peak/trough sign structure is independent of N ; the relative ordering of ζ and the random control at peaks does depend on N (at $N = 10^5$, ζ exceeds the control; at $N = 10^6$ the broader study of this program finds top-50-peak means $\bar{r} \approx 2.5$ for ζ versus $\bar{r} \approx 2.8$ for the control, the ordering reversing across $N_c \approx 1.2 \times 10^5$; see the companion phase-transition analysis).

TABLE 1. Decile-conditional mean $\bar{r}(q)$ of the inter-class energy ratio at $N = 10^5$ (5498 abscissae). Monotone in q : destructive at troughs, constructive at peaks. Regenerated by the notebook of Appendix A.

q	0.05	0.15	0.25	0.35	0.45	0.55	0.65	0.75	0.85	0.95
ζ	−0.998	−0.987	−0.950	−0.860	−0.738	−0.498	−0.169	+0.297	+0.770	+1.544
rand. mult.	−0.983	−0.939	−0.868	−0.762	−0.608	−0.386	−0.118	+0.275	+0.727	+1.406

The mechanism is transparent. A peak of $|D_F|^2 = E(1 + r)$ is, by definition, an event of anomalously large $|D_F|^2$; since the diagonal energy E is comparatively rigid, a large $|D_F|^2$ is achieved *precisely* by a large positive cross-term, i.e. $r > 0$. A trough is the reverse: an anomalously small $|D_F|^2$ at fixed E requires a large negative cross-term, $r < 0$. Constructive interference at peaks and destructive interference at troughs are thus two faces of the same identity $|D_F|^2 = E(1 + r)$, selected by the sign of the conditioning. Proposition 3.1 guarantees the two effects average to zero.

Remark 4.2. This also explains why an *unconditional* or weakly-conditioned estimate of r can land on either sign or near zero depending on the sampling window and the implicit weighting, which is the source of the discrepancy addressed next.

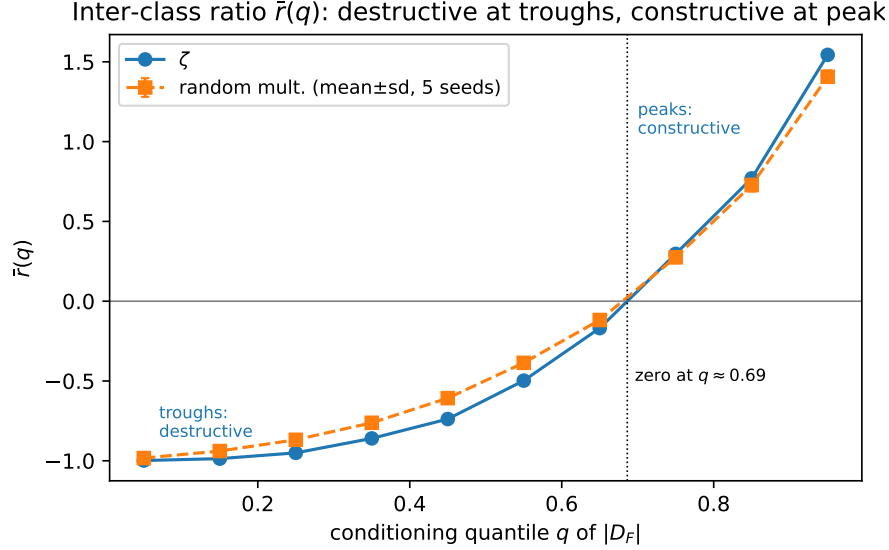


FIGURE 1. The decile-conditional mean $\bar{r}(q)$ of the inter-class energy ratio at $N = 10^5$, for ζ (circles) and the random multiplicative control (squares, mean $\pm$ s.d. over five seeds). The curve is monotone increasing, negative at troughs (low q), positive at peaks (high q), crossing zero near $q \approx 0.69$. By Proposition 3.1 the two regimes average to zero over all t ; the sign of r is therefore an attribute of the conditioning, not of ζ .

5. THE $t = 150000$ DISCREPANCY RESOLVED

The negative “gold-standard” value $r \approx -0.82$ that anchored the early destructive-interference reading was obtained at the specific abscissa $t = 150000$. We claim this point is a *trough*.

Proposition 5.1 (The anchor point is a trough). *At $N = 10^5$, the abscissa $t = 150000$ has local $|D_\zeta|$ -rank 0.12 within its neighbourhood (on the scale where 0 is the deepest trough and 1 the highest peak)—squarely in the lower tail—and correspondingly $r(150000; N) = -0.998$. The point is unambiguously a trough. The historically-cited value $r \approx -0.82$, obtained at the nearby scale $N = 1.5 \times 10^5$, is likewise a trough reading. Conditioning instead on the peaks of the same neighbourhood returns $\bar{r} > 0$, in accordance with Theorem 4.1.*

Thus the early measurement was not erroneous: it correctly reported the value of r at a trough. The error was only one of *interpretation*—reading a trough value as representative of ζ in general, when by Corollary 3.2 no such representative value exists. Once the conditioning is made explicit, the early report ($r < 0$ at a trough) and the later report ($r > 0$ at peaks) are not merely compatible: they are two predicted points on the single monotone curve $\bar{r}(q)$ of Theorem 4.1.

6. RECOMMENDED CONVENTION AND IMPLICATIONS

- (a) **Always specify the conditioning.** An inter-class interference statement of the form “ ζ exhibits destructive (resp. constructive) interference” is incomplete. The correct forms are “destructive *at troughs*” and “constructive *at peaks*.”
- (b) **Restatement of prior claims.** The early “cancellation-architecture” observations are correct as statements about troughs and low- $|D_F|$ events. The later phase-coordination observations are correct as statements about peaks. Neither is contradicted.

- (c) **Peaks are the RH-relevant regime.** Since the Riemann Hypothesis constrains the *growth* of $|\zeta(1/2 + it)|$ —i.e. its large values—the peak conditioning, where $r > 0$, is the one relevant to the hypothesis. The structural asymmetry between ζ and the random multiplicative control reported elsewhere in this program is correctly formulated at peaks.
- (d) **The monotone profile $\bar{r}(q)$ is itself the object of interest.** Rather than a single number, the full curve $\bar{r}(q)$ (negative at troughs, zero in bulk, positive at peaks) is the reproducible invariant; its slope and zero-crossing are the quantities worth comparing across function classes.

APPENDIX A. COMPUTATIONAL REPRODUCIBILITY

All quantitative claims in this note are reproduced by a single self-contained Python notebook, `P6_inter_class_ratio.ipynb` (also provided as `reproduce_colab.py`), runnable end-to-end in Google Colab with no external data. The notebook:

- (1) builds $\omega(n)$ for $n \leq N$ via a smallest-prime-factor sieve;
- (2) computes the class-sums $S_k(t; N)$ and $D_\zeta = \sum_k S_k$ with Kahan-compensated summation on a t -grid of spacing $2\pi/\log N$, and validates $D_\zeta(t; 10^4)$ against a 50-digit `mpmath` reference;
- (3) forms $r(t; N) = X/E$ and stratifies its conditional mean $\bar{r}(q)$ by the quantile of $|D_\zeta|$, producing the monotone curve of Theorem 4.1;
- (4) evaluates r and the local rank of $|D_\zeta|$ at $t = 150000$, confirming Proposition 5.1;
- (5) repeats for the random multiplicative control with fixed seeds $\{42, 137, 271, 314, 577\}$.

Default parameters ($N = 10^5$, a finely sampled window) run in a few minutes on a free Colab instance; the $N = 10^6$ configuration and the full $[10^5, 2 \times 10^5]$ trough census are provided as optional cells with stated runtimes. Random seeds are fixed; every figure and table in the note is regenerated by the notebook.

All numerical values reported in Section 4 and Section 5 (Table 1, Figure 1, and Proposition 5.1) are the verbatim output of this notebook at $N = 10^5$; they are reproduced bit-for-bit from the fixed seeds and require no external data.

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